



Artificial Neural Network (ANN) Hyperparameter Tuning and Performance Evaluation



SUBMITTED BY:
ASHWINI VISHAL HEBBALI
4MH23CA006

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Introduction

Artificial Neural Networks (ANNs) are widely used in machine learning to solve classification and prediction problems by learning complex patterns from data. However, the performance of an ANN depends significantly on the selection of hyperparameters such as the number of hidden layers, neurons, activation functions, weight initializers, optimizers, epochs, and batch size. This study aimed to systematically evaluate the impact of these hyperparameters on a binary classification problem and identify the optimal ANN configuration.

A series of experiments were conducted by modifying one parameter at a time while keeping the remaining architecture unchanged. The models were evaluated using validation accuracy, validation loss, training time, and other performance metrics. Based on the experimental results, an optimized ANN model was developed and evaluated on the test dataset.

Summary of Experiments

Task 1 – Hidden Layer Comparison

The first experiment investigated the effect of increasing the number of hidden layers. Four architectures containing **1, 2, 3, and 5 hidden layers** were evaluated. Although deeper networks achieved higher training accuracy, their validation accuracy gradually decreased while validation loss increased, indicating overfitting. The model with **one hidden layer** produced the highest validation accuracy (**86.12%**) and the lowest validation loss (**0.3497**), demonstrating superior generalization.

Task 2 – Number of Neurons

The second experiment analyzed the effect of changing the number of neurons in the hidden layer. Models with **4, 8, 16, 32, and 64 neurons** were compared. The model containing **32 neurons** achieved the highest validation accuracy (**86.37%**), while **64 neurons** resulted in the lowest validation loss. Considering both performance and computational complexity, **32 neurons** were selected as the optimal configuration.

Task 3 – Activation Function Comparison

Five activation functions (**ReLU, Sigmoid, Tanh, ELU, and Leaky ReLU**) were evaluated. Among these, **ReLU** achieved the highest validation accuracy (**86.69%**) and the lowest validation loss (**0.3448**). Although Tanh and ELU also produced competitive results, ReLU demonstrated better convergence and learning capability for the binary classification problem.

Task 4 – Weight Initializer Comparison

Different weight initialization techniques (**Uniform, He Uniform, He Normal, Glorot Uniform, Glorot Normal, and Random Normal**) were compared. The **Random Normal** initializer achieved the highest validation accuracy (**85.69%**), while **Glorot Uniform** produced the lowest initial loss. Considering overall predictive performance, **Random Normal** was selected as the preferred initializer.

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Task 5 – Output Layer Activation

Three output activation functions (**Sigmoid, Softmax, and Linear**) were studied. Although all three produced comparable numerical results, **Sigmoid** was identified as the most appropriate activation function because the dataset represents a **binary classification problem**. Softmax is designed for multiclass classification, while Linear activation is generally used for regression problems.

Task 6 – Optimizer Comparison

The performance of five optimization algorithms (**Adam, SGD, RMSprop, Adagrad, and Nadam**) was evaluated. The **Adam optimizer** achieved the highest validation accuracy (**85.75%**) and the lowest validation loss (**0.3557**). SGD trained the fastest but produced lower prediction accuracy. Therefore, Adam was selected as the most suitable optimizer.

Task 7 – Epoch Analysis

The model was trained using **10, 25, 50, 100, and 200 epochs**. The best performance was achieved at **25 epochs**, with a validation accuracy of **85.56%** and a validation loss of **0.3587**. Increasing the number of epochs beyond this point did not improve the model's performance and slightly increased validation loss, indicating the beginning of overfitting.

Task 8 – Batch Size Comparison

Different batch sizes (**8, 16, 32, 64, and 128**) were compared. Batch sizes **8 and 32** achieved the highest validation accuracy (**85.87%**). However, batch size **32** required significantly less training time and produced the lowest validation loss (**0.3547**). Therefore, batch size **32** was selected as the optimal value.

Task 9 – Improved ANN Model

An improved ANN model was developed using the best-performing hyperparameters identified during the experiments:

- **1 Hidden Layer**
- **32 Hidden Neurons**
- **ReLU Activation Function**
- **Random Normal Weight Initializer**
- **Adam Optimizer**
- **25 Epochs**
- **Batch Size = 32**
- **Dropout (0.2)**

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The optimized ANN achieved a **Test Accuracy of 86.25%** with a **Test Loss of 0.3385**, demonstrating effective learning and good generalization. The addition of the Dropout layer helped reduce overfitting and improved the model's robustness.

Task 10 – Final Model Evaluation

The final ANN model was evaluated using standard classification metrics, including **Accuracy, Precision, Recall, F1-Score, Classification Report, and Confusion Matrix**. These metrics provide a comprehensive assessment of the model's performance by measuring not only overall prediction accuracy but also the balance between correctly identifying positive and negative classes. The confusion matrix further illustrates the distribution of correct and incorrect predictions, providing valuable insight into the classifier's effectiveness.

Overall Conclusion

This project demonstrated the importance of systematic hyperparameter tuning in improving the performance of Artificial Neural Networks. Each experiment highlighted how different architectural and training parameters influence model accuracy, loss, convergence, and computational efficiency. The study showed that increasing network complexity does not necessarily improve predictive performance; instead, selecting appropriate hyperparameters leads to better generalization.

Among all the experiments, the optimal ANN configuration consisted of **one hidden layer with 32 neurons, ReLU activation, Random Normal weight initialization, the Adam optimizer, 25 training epochs**, and a **batch size of 32**, along with a **Dropout layer** to reduce overfitting. This optimized model achieved a **test accuracy of 86.25%** while maintaining a low test loss of **0.3385**.

Overall, the experiments demonstrate that careful hyperparameter optimization is essential for developing efficient and reliable neural network models. The final ANN model successfully balances prediction accuracy, generalization capability, and computational efficiency, making it suitable for binary classification tasks and providing a strong foundation for future machine learning applications.